Spider Fusion in Qutrit Quantum Circuits: A ZX-Calculus Study of Diagrammatic and Circuit-Level Optimization

Hannah Reynen

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1 Introduction and Motivation

Quantum circuit optimization is a key element of the quantum software stack, as improvements in metrics such as gate count and circuit depth directly translate to more efficient hardware implementations. In the current era of quantum computing, hardware constraints remain a primary limitation. As such, the development of effective optimization techniques is especially important. One such framework is ZX-calculus, which provides a graphical representation of quantum circuits and can reveal simplifications that are not always apparent in typical circuit models. While ZX-calculus has been extensively studied for qubit-based systems, leading to optimization tools such as PyZX, its application to higher-dimensional qudit systems remains comparatively underexplored.

In this project, I explore 3-level (qutrit) systems in particular, as they naturally occur in physical implementations such as neutrino systems or offer advantages in certain quantum algorithms, such as QAOA [3]. Spider fusion is a ZX rewrite rule that merges adjacent spiders of the same kind, thereby reducing diagram complexity and advancing the goal of this project: to investigate whether diagrammatic simplification in the qutrit ZX framework translates to meaningful circuit-level optimizations. To explore this question, I construct several ZX graphs representative of qutrit circuits for instances such as the Quantum Fourier Transform (QFT) and Trotterized Hamiltonians, and then apply automated Z-spider fusion rules. The resulting simplified diagrams are then analyzed to determine the extent to which fusion leads to reductions in structural complexity and circuit resources.

2 Methods and Model Assumptions

To study the effect of spider fusion simplifications on qutrit circuits, I represent each circuit as a graph using Python's NetworkX multigraph tool [5]. In this implementation, each node corresponds to a spider (Z, X, or boundary), and edges represent the wires connecting them. Each spider carries a phase parameter, and the graph's connectivity encodes the quantum circuit's underlying structure.

2.1 Qutrit Background

Standard qubit circuits use a two-dimensional computational basis, while qutrit circuits use a three-dimensional basis:

$$\text{qubit: } \{|0\rangle, |1\rangle\}, \qquad \text{qutrit: } \{|0\rangle, |1\rangle, |2\rangle\}$$

This changes the algebra of the circuit representation. For qubits, the Pauli Z and X operators are:

$$Z_2 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad X_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

For qutrits, these operators generalize to:

$$Z_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}, \qquad X_3 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad \omega = e^{2\pi i/3}$$

Here, Z_3 applies phase factors determined by the third root of unity, while X_3 cyclically shifts the computational basis states according to $|\, j\, \rangle \mapsto |\, j + 1 \mod 3\, \rangle$. The qutrit analogue of the Hadamard gate is the three-dimensional Fourier transform:

$$F_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}$$

These definitions establish the qutrit-native operator framework used throughout this project, where circuit structure and phase behavior are represented diagrammatically using qutrit ZX-calculus.

2.2 Z-Spider Fusion Rule

From the ZX-Calculus rewrite rules, the sole rule implemented in this study is the spider fusion rule. When two Z or X spiders are connected by a plain edge (as opposed to a Hadamard edge), they can be merged into a single spider whose phase is the sum of the original spiders' phases (modulo 3). Diagrammatically, this corresponds to replacing two adjacent spiders with one whilst preserving all external wire connections.

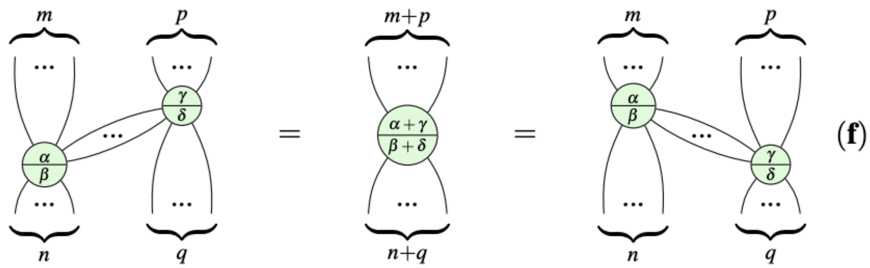


Figure 1: Z- and X-spider fusion rules in qudit ZX-calculus. Two connected spiders of the same type merge into a single spider with a sum of phases (modulo 3). Adapted from **[1]**

Formally, if two Z-spiders with k and m legs and phases p_1 and p_2 are connected, the fusion rule is:

$$Z_{p_1}^{(k)} \;-\; Z_{p_2}^{(m)} \;\longrightarrow\; Z_{(p_1+p_2) \bmod 3}^{(k+m-2)} \qquad (2)$$

In the graph implementation, this rule is applied by identifying two Z-spiders connected by a plain edge, summing their phases, rewiring so the total number of inputs and outputs on the new merged node is equivalent to that of the previous separate nodes, and deleting the redundant spider. This process is then repeated iteratively until no further ZZ pairs can be reduced.

2.3 Trotterization and Strang Splitting

To generate additional test instances, Hamiltonian time evolution is approximated using Trotterization. For a Hamiltonian of the form

$$H = A + B \qquad (3)$$

the exact time evolution is given by

$$U(t) = e^{-i(A+B)t} \qquad (4)$$

In the test cases, A and B are chosen such that they do not generally commute, allowing for the time evolution to be approximated using second-order Strang splitting:

$$U(t) \approx \left(e^{-iAt/(2r)}\, e^{-iBt/r}\, e^{-iAt/(2r)}\right)^r \qquad (5)$$

This symmetric decomposition improves approximation accuracy by scaling the error from $O(t^2/r)$ in the first-order implementation to $O(t^3/r^2)$ in the second-order implementation. The alternating Z and X-type spiders naturally introduce a repeated-phase structure, making them well-suited for testing ZX-fusion rules.

The Hamiltonians considered are as follows:

$$H = \alpha Z + \beta X \qquad \text{(non-entangling)} \qquad (6)$$

$$H = \alpha(Z \otimes Z) + \beta(X \otimes I) \qquad \text{(entangling)} \qquad (7)$$

2.4 Test Circuits and Metrics

Three classes of circuits were analyzed: qutrit Quantum Fourier Transform (QFT) circuits for system sizes up to n=20, and Trotterized evolutions of both non-entangling and entangling Hamiltonians. Additional small instances were used for validation purposes.

Diagrammatic complexity was quantified using graph-based metrics, including the number of Z-spiders, the number of X-spiders, the total number of nodes, the total number of edges, and the number of fusion operations applied. These metrics provide a proxy for circuit complexity, as reductions in the ZX graph are expected to correspond to reductions in circuit-level resources such as gate count or depth, even though an explicit circuit extraction step is not performed in this work.

3 Results and Validation

This section presents the results of applying Z-spider fusion to several classes of qutrit circuits, along with validation checks to ensure the correctness of the implementation. For each test case, both diagrammatic complexity and qualitative structural changes are analyzed to assess whether ZX-based simplifications correspond to actual circuit-level improvements.

3.1 Quantum Fourier Transform

For the QFT case, a significant reduction in diagram complexity was observed. As shown in the table, as the number of qutrits increased, the number of initial Z-spiders also increased; yet after the Z-spider fusion rule was applied, the diagram collapsed to a single Z-spider in all cases, regardless of system size (see Table 1).

Table 1: Z-Spider Reduction Under Fusion for Qutrit QFT Circuits

Qutrits (n)	Z-Spiders (Before)	Z-Spiders (After)	Number of Fusions
3	6	1	5
5	20	1	19

This result indicates that the ZX representation of QFT is largely composed of fusion-compatible phase structure,

8	56	1	55
10	90	1	89
15	210	1	209
20	380	1	379

leading to these massive simplifications. However, this reduction does not translate into a meaningful circuit-level resource reduction. Instead, the fusion collapses the distributed phases and entangling structure of QFT into a single Z-spider, concealing the original gate-level structure of the circuit. This result highlights an important distinction between diagrammatic simplification and circuit optimization. While ZX fusion can significantly reduce the structural composition, additional rewrites or extraction methods are needed to determine a useful circuit-level representation.

3.2 Non-entangling Hamiltonian

For the non-entangling Hamiltonian $H = \alpha Z + \beta X$, significant reductions in Z-spiders and overall nodes were observed. As Trotter steps increased to r=20, the percentage of Z-node reductions approached 50%, and the percentage of total node reductions reached 30%. As shown in Figure 2, the Z-spiders and total nodes scale linearly with the number of Trotter steps, both in the initial and simplified instances. However, in both cases, the diagram's slope is reduced. This indicates that a considerable amount of Z-spider phase structure is eliminated. Particularly, the number of Z-spiders was reduced from 2r to r+1, showing that the repeated phases introduced by the Strang Trotterization are highly compatible with Z-spider fusion.

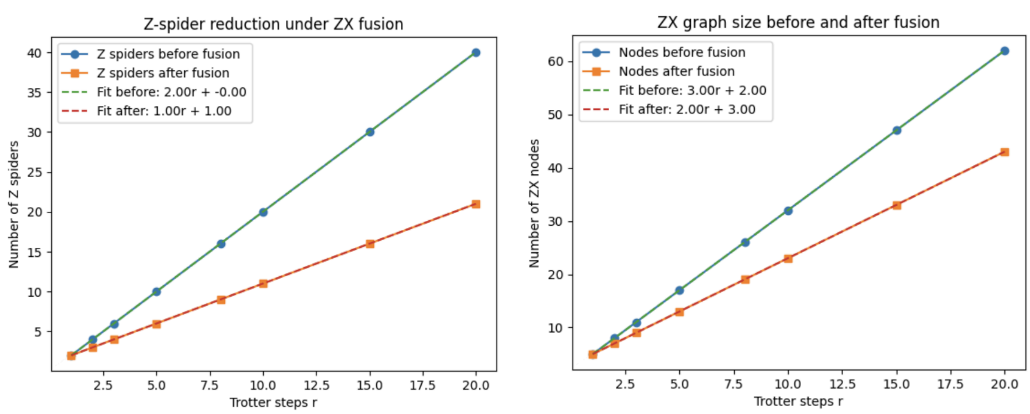
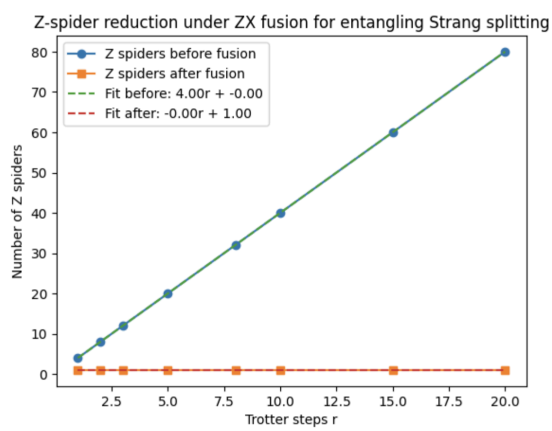


Figure 2: Z-spider and node counts vs. Trotter steps for the non-entangling Hamiltonian before and after fusion.

These results suggest that Z-spider fusion is especially effective at compressing repeated phase structure in circuits generated by Trotterized evolution. Because Strang splitting creates a symmetric sequence of alternating Z and X evolutions, the resulting ZX representation contains repeated adjacent Z-spiders. These repeated phase structures are highly compatible with the fusion rule, leading to consistent reductions in diagram complexity. Importantly, unlike in the QFT case, this simplification preserves the overall circuit structure while reducing redundancy. Thus, for non-entangling Hamiltonians, ZX fusion corresponds more directly to meaningful circuit-level simplifications, as redundant phase operations are systematically combined.

3.3 Entangling Hamiltonian

For the entangling Hamiltonian, $H=\alpha(Z\otimes Z) + \beta(X\otimes I)$, Strang splitting is also implemented to construct the Trotterized circuits. As shown in the figures, both the number of Z-spiders and the total number of nodes scale linearly with the number of Trotter steps. After applying Z-spider fusion, a significant reduction in diagrammatic complexity is observed. Specifically, the number of Z-spiders reduces from 4r to a constant 1 across all tested Trotter steps r, as seen in Figure 3. This shows that all the Z-spiders in the entangling Hamiltonian reduce to a single Z-spider, regardless of system size. Looking at the total node count, there is also a large reduction from 5r+4 total original nodes to 1r+5 nodes after simplification. This decrease in the total number of nodes signifies the elimination of repeated phase structures while the X structure remains intact.



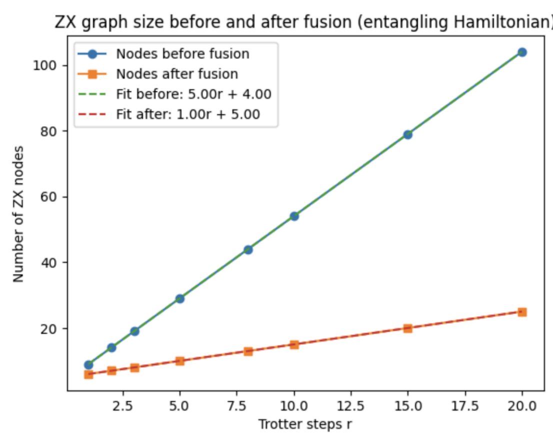


Figure 3: Z-spider and node counts vs. Trotter steps for the entangling Hamiltonian before and after fusion.

As in the QFT case, this strong diagrammatic compression does not correspond to a reduction in circuit resources. The fusion of Z-spiders collapses the entangling interactions to a single Z-spider with a global phase, concealing the original entangling interactions between the qutrits. As a result, while the ZX diagram becomes significantly smaller, the simplifications do not preserve a meaningful gate-level interpretation of the original entangling circuit. Further, these results emphasize that Z-spider fusion alone only captures a limited part of the circuit structure, the adjacent phase redundancy, and thus is insufficient for simplifying entangling operations without the use of additional ZX rewrite rules.

3.4 Validation Checks

To ensure the implementation was correct, several validation checks were performed. These include structural verification of constructed circuits, numerical validation of the fusion rule, and convergence checks for Trotterized Hamiltonian evolution.

3.4.1 Quantum Fourier Transform Case

The correctness of the ZX-based QFT representation was validated through both structural and numerical checks. Structurally, for small system sizes (n=2,3,4,5,6), each constructed graph was verified to contain n boundary nodes, n(n−1) Z-spiders, and n Hadamard edges, consistent with the expected global properties of the n-qutrit QFT. In addition, for each qutrit wire, the number of incident Z-spider endpoints was confirmed to be n−1, consistent with the fact that each wire participates in pairwise phase interactions with every other wire.

The implemented Z-spider fusion rule was numerically validated by representing each spider as a qutrit tensor, contracting two connected spiders along their shared edge, and comparing the result to the tensor of the fused spider with phase addition modulo 3. The resulting tensors agreed to within numerical precision (~10-15), with both the Frobenius norm difference and the maximum absolute error equal to zero. This confirms that the fusion rule preserves the linear map represented.

3.4.2 Hamiltonian Cases

For each Hamiltonian case, I verified that my Trotterized implementation converged to the exact time-evolution unitary, up to a global phase, as the number of Trotter steps increased.

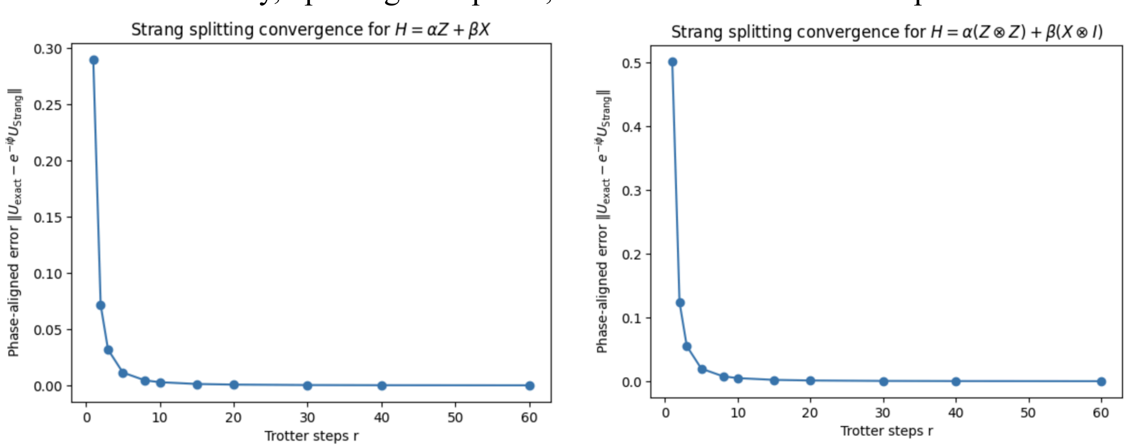


Figure 4: Error in the Trotterized unitary compared to the exact evolution as a function of Trotter steps.

3.4.3 Controlled Redundancy Tests

To test the behavior of the simplification procedure under both fusion-compatible and non-compatible modifications, two controlled test instances were constructed based on the previously defined Trotterized Hamiltonian circuits. In the first test, fusion-compatible redundancy was introduced by adding additional zero-phase Z-spiders. This preserves the implemented unitary while increasing diagram complexity. Applying the automated fusion routine reduces the obfuscated diagram not only by removing the introduced redundancy, but also by simplifying pre-existing fusion-compatible structure. As a result, the final diagram is smaller than both the obfuscated and original instances. This demonstrates that the method can remove deliberately inserted redundancy and further simplify the original circuit.

Table 2: Metrics for Base, Redundant, and Simplified Diagrams

Case	Nodes	Z-Spiders	X-Spiders	Edges	Boundary
Base	11	6	3	10	2
Redundant	13	8	3	12	2
Simplified	9	4	3	8	2

In a second test, an additional non-fusion-compatible structure was inserted between the two qutrit wires using intermediate X-spiders. This increased the graph size but did not add new Z-Z adjacencies, so the fusion rule could not eliminate it. The resulting graph, therefore, remained larger after simplification. These tests confirm that the implementation behaves consistently with its design: it removes fusion-compatible redundancy but does not act as an all-purpose optimizer for arbitrary circuit structure.

Table 3: Metrics for Baseline and Negative-Control Circuits Before and After Fusion.

Test Case	Stage	Nodes	Z-Spiders	X-Spiders	Edges	Boundary
Baseline	Before	19	12	3	23	4
Baseline	After	8	1	3	10	4
Negative Control	Before	21	12	5	27	4
Negative Control	After	10	1	5	14	4

4 Full Stack Impact and Resource Implications

By studying the pipeline from algorithm to circuit to ZX diagram to rewritten diagram, we can examine the impact of ZX rewrite rules on circuit resources. The baseline for comparison is the original ZX graph, obtained by directly converting the original qutrit circuit to a ZX diagram before applying any rewrite rules. After applying the ZX-spider fusion rewrites, the diagram size is measured through spider counts and total node counts.

The greatest resource-level improvement was in the non-entangling Hamiltonian example: $H = \alpha Z + \beta X$, where ZX fusion merges consecutive phase spiders introduced by Strang splitting. For larger Trotter step counts, the rewrite reduces the total ZX node count by 30%. This corresponds directly to a reduction in circuit operations as the constructed ZX diagram represents sequential operations of Z and X gates. This shows that ZX rewrites can produce a meaningful reduction in circuit resources by simplifying Trotterized circuits that are largely composed of local phase rotations.

In contrast, the entangling Hamiltonian and QFT examples show significant diagrammatic compression, but limited circuit-level reductions. In the entangling Hamiltonian, Z-spider fusion merges redundant phase structure, but does not remove the entangling connectivity of the $Z \otimes Z$ interaction. Similarly, in the QFT example, many Z-spiders collapse into a single fused spider, revealing global phase structure, but the underlying controlled-phase connectivity remains.

Thus, these results suggest that ZX-spider fusion is most effective for simplifying chains of phase operations, but less effective for circuits dominated by entangling topology, which only see large structural compression rather than actual gate reductions.

5 Limitations & Future Work

The main limitation of this project is that only Z-spider fusion rules were implemented. Beyond spider fusion, ZX-calculus includes additional rewrite rules that can reduce diagram size or expose simplifications, such as identity removal, color-change-related manipulations, and bialgebra/Hopf-style interactions. However, for the qutrit circuit families studied here, spider fusion is the most consistently applicable rule and the one that produced the clearest measurable reductions in diagram resources. Furthermore, this study concentrates on a limited number of circuit classes and does not explore the full range of possible qutrit circuit structures.

In addition, this project does not perform full circuit extractions of the simplified ZX diagrams. As such, reductions in diagram complexity do not necessarily correspond to reductions in circuit resources such as gate count or circuit depth, especially in entangling cases where spider fusion leads to overcompression of the diagram structure.

Thus, this analysis focuses solely on graph-based metrics and does not directly examine circuit resources. Graph-based metrics are used as proxies for circuit complexity, providing insight into structural reductions but not directly quantifying hardware-level resources. With this, the qutrit/qudit ZX framework is also less well studied than the qubit cases, which limits the availability of resources and benchmarking tools, hindering the scope of optimization in this project.

The limitations of this project inform avenues for future work. Using additional ZX rewrite rules will allow for further simplifications and for a larger range of circuit classes, particularly those with entangling structure. Another route for future work would be to develop a circuit-extraction procedure that directly shows how structural reductions affect circuit-level resources and, in turn, hardware performance. Future work might also include benchmarking these simplifications against an existing ZX framework, such as comparing the qubit case using compilers like PyZX. Another worthwhile exploration would be in larger, more complex algorithmic circuits to see if these reductions persist in realistic quantum computing applications.

6 Resources

[1] Alex Townsend-Teague & Konstantinos Meichanetzidis (2021): *Simplification Strategies for the Qutrit ZX-Calculus*. arXiv preprint arXiv:2103.06914. Available at https://arxiv.org/abs/2103.06914.

[2] Daniele Trisciani, Marco Cattaneo & Zoltán Zimborás (2025): *Decomposition of multi-qutrit gates generated by Weyl-Heisenberg strings*. arXiv preprint arXiv:2507.09781. Available at https://arxiv.org/abs/2507.09781.

[3] Gabriel Bottrill, Mudit Pandey & Olivia Di Matteo (2023): *Exploring the Potential of Qutrits for Quantum Optimization of Graph Coloring*. arXiv preprint arXiv:2308.08050. Available at https://arxiv.org/abs/2308.08050.

[4] John van de Wetering (2020): *ZX-calculus for the working quantum computer scientist*. arXiv preprint arXiv:2012.13966. Available at https://arxiv.org/abs/2012.13966.

[5] NetworkX Developers (2025): NetworkX (Python package) Documentation. Available at https://networkx.org/documentation/stable/reference/index.html.

[6] Quanlong Wang (2018): *Qutrit ZX-calculus is Complete for Stabilizer Quantum Mechanics*. arXiv preprint arXiv:1803.00696. Available at https://arxiv.org/abs/1803.00696.

[7] Shruti Dogra, Arvind & Kavita Dorai (2015): *Implementation of the quantum Fourier transform on a hybrid qubit-qutrit NMR quantum emulator*. arXiv preprint arXiv:1503.06624. Available at https://arxiv.org/abs/1503.06624.

7 Appendix: GitHub Repository

The code, final report, and reproducible notebook for this project are available at:
https://github.com/hrnewmoon/Physics-765-Final-Project